Project Part 1

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Team GitHub Repository: https://github.com/tksoftw/stat-3080-project-1

Introduction

Capital Bikeshare is a bike-share service prevalent in Washington, D.C. and the surrounding metropolitan counties. As bike-sharing ridership grows in densely populated areas, questions arise about whether these stations correlate with local crime patterns. Companies, urban planners, and residents would benefit from knowing whether Capital Bikeshare locations are associated with safer or less-safe surroundings.

Our research question is: Does proximity to a Capital Bikeshare station correlate with higher rates of property crime but lower rates of violent crime in Washington, D.C. in 2026? We examine two crime categories separately because the theoretical mechanisms differ: property crime may be attracted to high-foot-traffic hubs, while violent crime may be deterred by increased pedestrian activity and natural surveillance.

Data Summary

We use two open datasets from the D.C. government, scoped to Washington, D.C. in 2026. Because they include all reported incidents and all active stations in 2026, they represent population data for reported crimes.

- **Capital Bikeshare Locations**: Contains 819 active stations, with `LATITUDE` and `LONGITUDE` coordinates.
- **Crime Incidents in 2026**: Contains 4,801 reported offenses with location data and `OFFENSE` category.

A crime incident is labeled "near" a bikeshare station if it falls within approximately 310 meters, or a 5-block radius, of any station. Otherwise, it is "far". Incidents are also classified as "Property" such as theft and burglary, or "Violent" such as robbery and homicide.

Data Limitations: The dataset contains only *reported* crimes; minor property thefts are likely underreported, especially in certain neighborhoods. Furthermore, the GPS coordinates of incidents represent approximate block-level locations rather than exact addresses, introducing some measurement error in the distance calculations. Additionally, some Capital Bikeshare locations were located far outside D.C. in the surrounding suburbs, so we filtered out any stations outside a 12 km radius from the geographic center of D.C. Despite these limitations, the data are appropriate to answer our research question because they represent the most comprehensive geographic record of urban crime and transit behavior available for our scope.

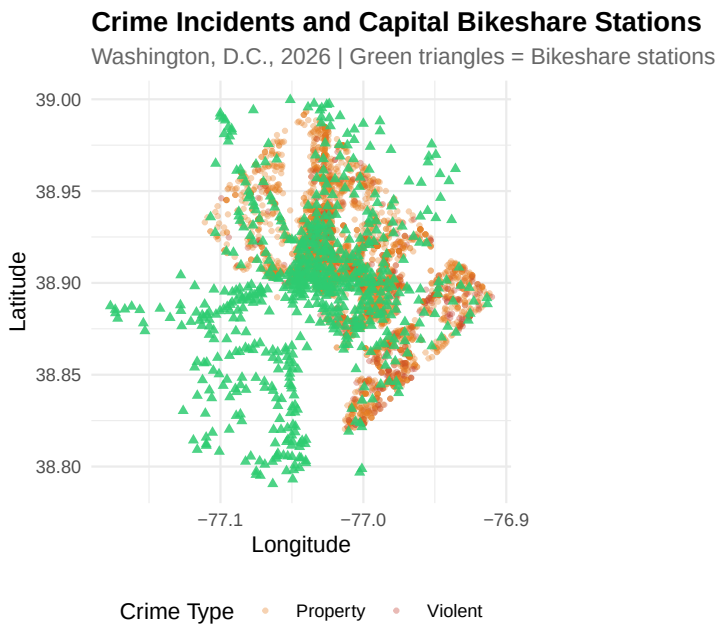
Exploratory Analysis

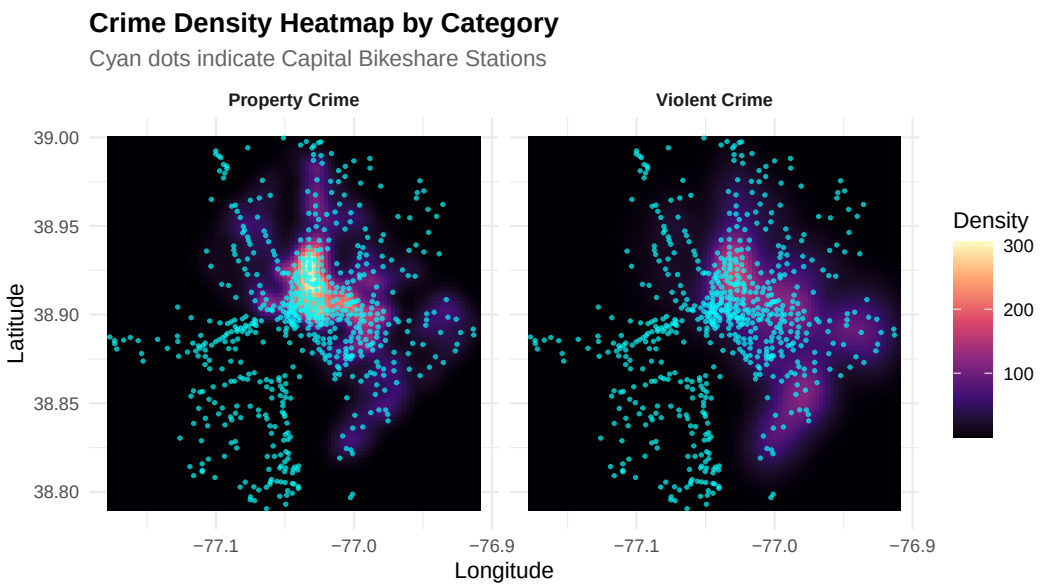
```
## Numerical Summary 1: Incident Counts by Proximity and Crime Category

##          Category
## Proximity Property Violent
##      Far       908     185
##      Near     3295     413

## Numerical Summary 2: Crime Category Proportions Within Proximity Group (%)

##          Category
## Proximity Property Violent
##      Far      83.1    16.9
##      Near     88.9    11.1
```





Conclusions

The numerical summaries reveal that of 4,801 total incidents, property crimes dominate at 88.9% near bikeshare stations vs 83.1% far from them. Conversely, violent crimes are proportionally lower near stations at 11.1% compared to 16.9% further away.

The crime dot map visualizes why this occurs: bikeshare stations, represented as green triangles, are densely concentrated in D.C.'s core, where orange property crime dots heavily

cluster. The density heatmap makes this even clearer. Both crime types share a similar geographic core in central D.C., but the property crime hotspot is more intense and tightly aligned with the bikeshare network. Violent crime, by contrast, is more diffuse across the city, including in areas with fewer stations. The visual patterns thus suggest a difference in crime composition depending on proximity to bikeshare stations.

References

Crime Incidents in 2026. (n.d.). [Dataset]. Link to source

Capital Bikeshare locations. (n.d.). Link to source

Rimolo, S. (2023, April 20). Bicycle share systems: a predictor of crime? Link to source

Google. (2026). Gemini (March 8 version) [Large language model]. Link to source